Proposed Solution

Phase 3: Project Design Phase

1. Solution Summary

trvale-book is a web application that converts a traveller's preferences into an AI-generated, editable and budget-checked itinerary. The user interacts with a React frontend. A FastAPI backend validates the request, coordinates the AI call and calculates the budget. Supabase provides authentication and the database. The interface then provides actions to review, edit, save, delete and print the plan.

2. User Workflow

1. Open the deployed site and choose Sign In, Sign Up or the one-click demo sign-in.
2. Start a new plan from the dashboard.
3. Complete the five wizard steps: destination and dates; travellers and budget; style and interests; dietary and requirements; final review.
4. Generate the itinerary and wait for the loading state to finish.
5. Review the day tabs, activity details and the budget summary.
6. Edit or delete activities, or regenerate a day, if required.
7. Open Saved Trips to search, filter, reopen or delete plans.
8. Print the plan or choose Save as PDF in the browser print dialog.

3. AI Generation

The backend forwards the validated request to the AI service, which builds a prompt that requests a fixed JSON structure with days, accommodation, transit tips and activities. The response is parsed and each day and activity receives an identifier and default values for missing fields. If no provider key is configured, or the provider call fails, the heuristic generator produces a plan so that the flow completes.

4. Budget Calculation

For each trip the budget service adds accommodation costs and activity costs. Activities in the food category, or with a meal type, count as food; transit and transport activities count as transit; everything else counts as activities. The variance equals the target budget minus the estimated total. A warning appears when the plan is over budget, or when more than forty percent of the budget remains unused.

5. Persistence and Saved Trips

The trips endpoints list, create, read, update and delete trips and enforce ownership for signed-in users. When Supabase is configured, the list endpoint reads the user's trips with their days and activities, and the create endpoint inserts the trip record. In demo mode, trips are stored in server memory and in the browser's localStorage so that the application remains usable without a database.

6. Security Model

The backend validates the Supabase bearer token for signed-in users. Row-level security in PostgreSQL restricts every table to its owner. Demo mode is controlled by the ENABLE_DEMO_MODE setting and should be disabled for a production deployment with real users.

7. Deployment Model

The frontend is hosted on Vercel and the backend is prepared for Render. The health endpoint reports the status, version, environment and whether Supabase and an AI provider are configured, which makes it a useful deployment smoke test.

8. Expected Benefits and Limitations

- A consistent starting point for planning with structured daily activities.
- A live budget comparison and warnings.
- Editable, savable and printable plans.

- A zero-setup demo mode for evaluation.

AI-generated plans require human review and all costs are estimates. Booking, payment, live weather and real-time provider data are not implemented.